

social media and other such options.

Also, most importantly, you would need a technology expert, preferably a full-stack one to exploit hardware and software aspects of the system. These skillset requirements are the minimum. If the application and systems are too complicated, then it will make sense to hire specialists for individual elements and have a mixed team of experts.

Top five red team skills

The most important skill any of the red team members can have is to think as negatively as possible and remain as creative as they can be when executing the job.

Creative negative thinking. It is the core goal to continually find new tools and techniques to invade systems and eventually protect the organisation's security. Moreover, showing flaws in the system needs a level of negative thinking to counter the inbuilt optimism in an individual.

Profound system-wide knowledge. It is imperative for red team members to have a deep understanding of computer systems, hardware and software alike. They should also be aware of typical design patterns in use.

Additionally, this knowledge should not be limited to only computer systems. It must span across many systems to involve heterogeneous systems.

Penetration testing (or pen testing). This is a common and fundamental requirement in the cybersecurity world. Moreover, for red teams, it becomes an essential part and a kind of standard procedure.

Software and hardware development. Having this skill means, as a red team member you can envisage and develop tools required to execute tasks. Moreover, knowing how AI systems are designed in the first place means you are likely to know failure points. One of the critical risks AI systems always pose is logical errors. These errors do not break the system but make the system behave in a certain way, which is not intended or may cause more damage.

If a red team member has experience in software and hardware development, then it is highly likely that he or she has seen all typical logical errors, and can, hence, exploit these

to accomplish the job.

Social engineering. This goes without saying as manipulation of people into doing something that can lead the red team to its goal is essential. The people aspect is also one of the failure vectors that the actual attacker would use. **Human errors and mistakes are one of the most frequent reasons for cyber problems.**

Inhouse or outsourced?

The next key question is: should you hire your team members and employ inhouse or outsource the red team activity?

We all know that security is one of those aspects where funding is mostly dry. It is tough because ROI on security initiatives cannot be proven easily. Unless something goes wrong and is prevented, it is nearly impossible to visualise or imagine. This limitation makes it difficult to convince that investment in security is worth making.

A quick answer to an inhouse or outsourced question would be: it depends on company size and budget.

If you have deployed AI systems for long-term objectives, then the inhouse red team would be the right choice as it would be engaged continuously. However, that comes with an additional ongoing budget overhead.

On the contrary, if you are unsure about the overall outlook, outsourcing is a better way to start. This way, you can test your business case for inhouse hiring in the long run.

From privacy and stricter controls perspective, the inhouse red team is highly justifiable. Red and blue teams' activities are more like cat-and-mouse games. When done correctly, each round can improve and add to the skillset of both teams, which, in turn, would enhance the organisation's security.

You can utilise the outsourcing option if you are planning to run a more extensive simulation. If you need specialised help or are looking for a specific skillset for a particular strategy execution, then also this option would make sense.

Objectives of a red team

Primarily, the red team exists to break the AI system and attached processes



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